

CUSTOMER CHURN ANALYSIS

Data Analytics Mini Project Report

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Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Power BI, Power Query and DAX

Project Objective

Analyze customer demographic, service, contract, billing and churn information to identify patterns associated with customer churn and retention. The project combines Python-based data cleaning and exploratory analysis with an interactive Power BI dashboard.

1. Dataset Overview

The cleaned dataset contains **1,200 customer records** and 23 columns. It includes demographics, tenure, subscribed services, contract type, payment method, monthly charges, total charges and churn status. Two analytical fields - Tenure_Group and Churn_Flag - were created during data preparation.

Metric	Value
Total customers	1,200
Churned customers	501
Retained customers	699
Overall churn rate	41.75%
Average monthly charges	78.87
Average total charges	2,744.84
Average tenure	35.21 months

2. Data Loading, Exploration and Cleaning

The Python workflow loads the raw CSV with Pandas, checks the first and last records, dataset shape, column names, data types, descriptive statistics, missing values, duplicates and unique values. Exact duplicate rows are removed and leading/trailing spaces are stripped from text fields.

Tenure_Months, Monthly_Charges and Total_Charges are converted to numeric values. Missing tenure and monthly-charge values are filled with their medians, while missing Total_Charges values are estimated as Monthly_Charges x Tenure_Months. Missing categorical values, if present, are filled using the mode.

Tenure is grouped into 0-12, 13-24, 25-48, 49-60 and 61+ months. A Churn_Flag field is created where 1 represents a churned customer and 0 represents a retained customer. The cleaned dataset is then exported for Power BI.

3. Python Analysis and Visualizations

The analysis calculates overall KPIs, customer distributions, churn rates by major categories, monthly-charge and tenure comparisons, top customers by total charges, higher-churn segments and major factors associated with churn. Fourteen visualizations were produced.

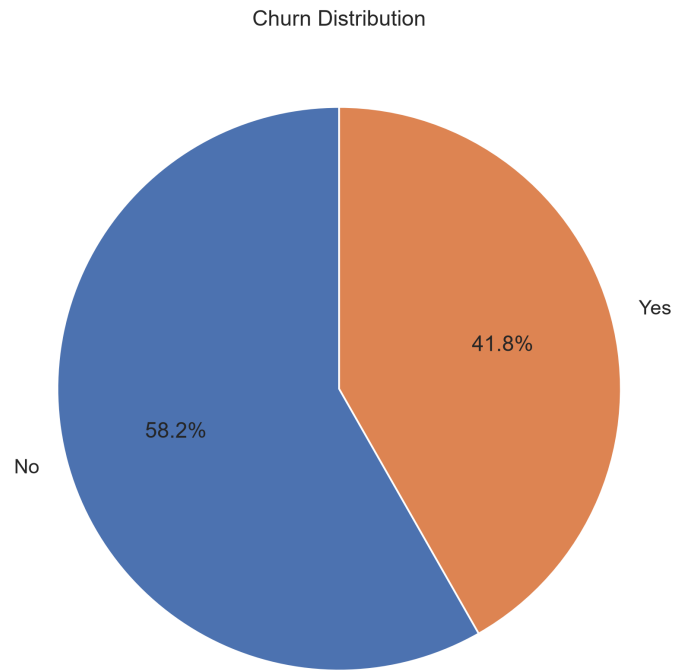


Figure 1. Overall churn distribution.

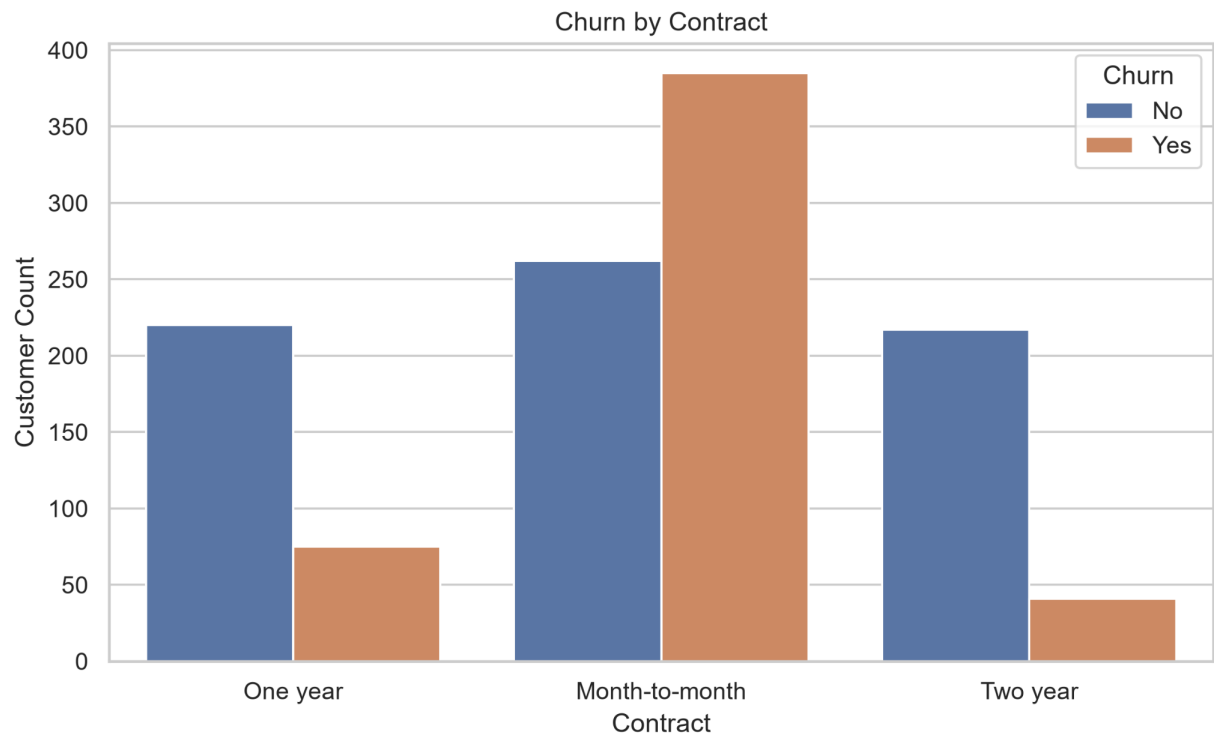


Figure 2. Churn by contract type.

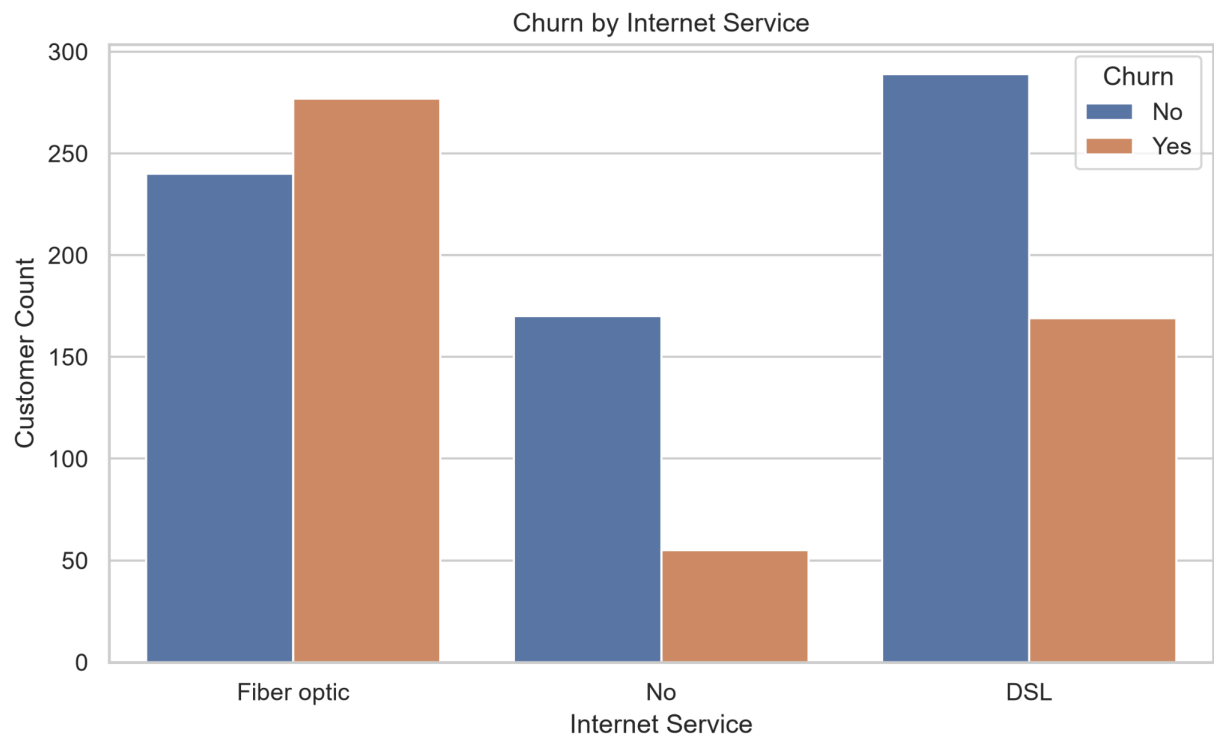


Figure 3. Churn by internet service.

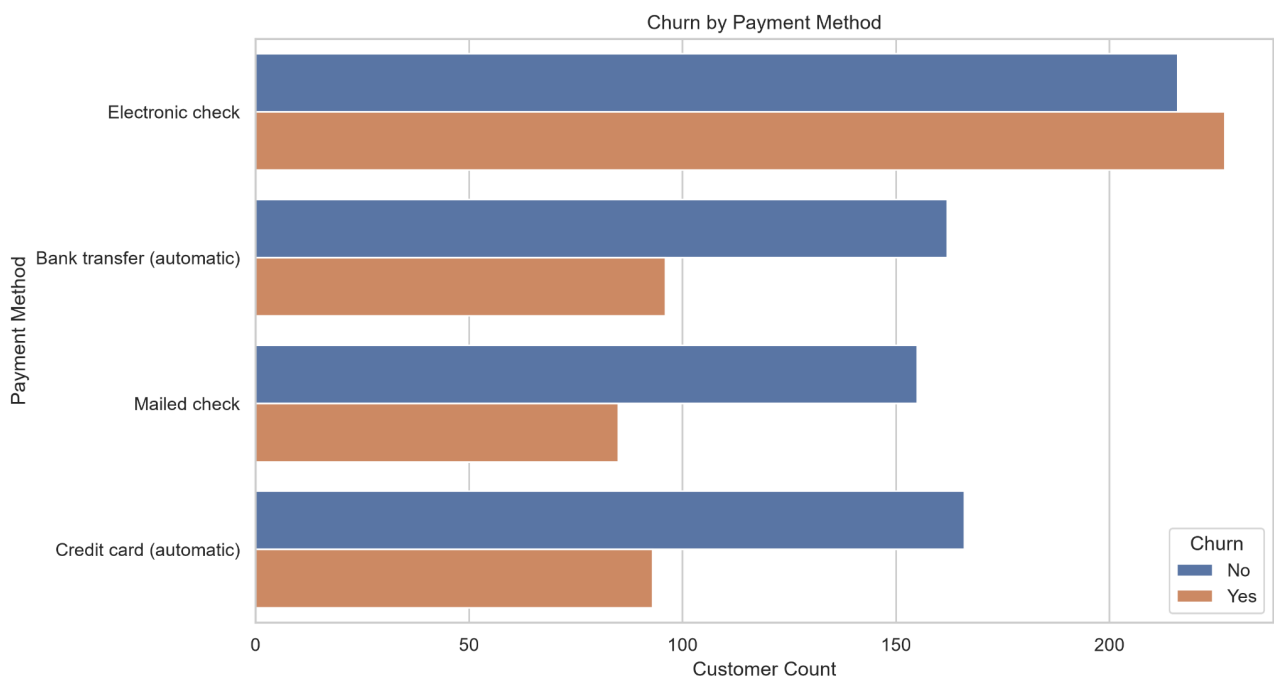


Figure 4. Churn by payment method.

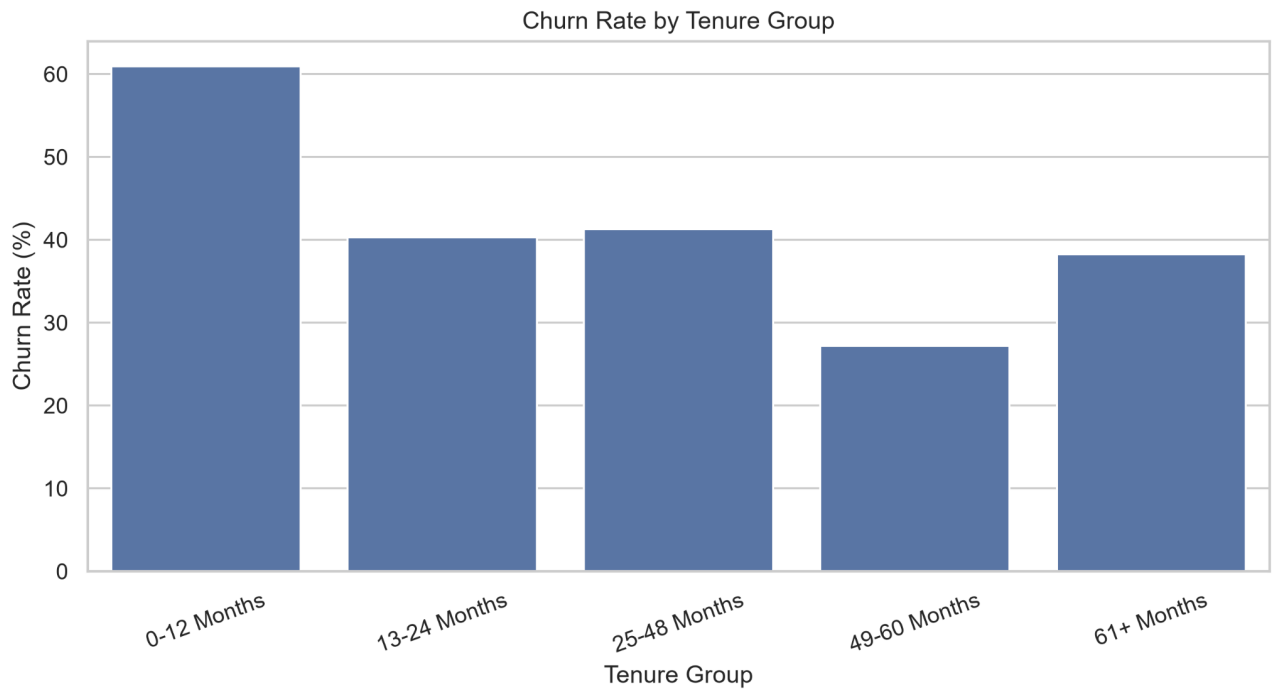


Figure 5. Churn rate by tenure group.

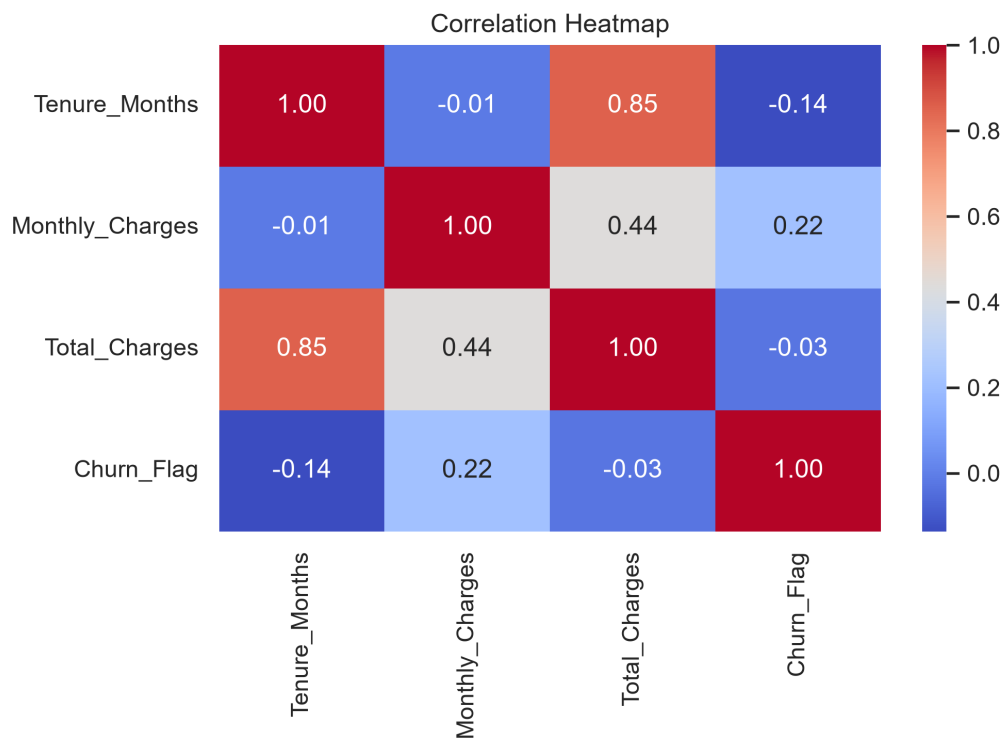


Figure 6. Correlation heatmap for key numeric variables.

4. Power BI Dashboard

The Power BI dashboard is designed as a single-page Customer Churn & Retention Analytics Dashboard. It contains KPI cards for Total Customers, Churned Customers, Retained Customers, Churn Rate %, Average Monthly Charges and Average Tenure. Visuals cover churn distribution, churn rate by contract, internet service and payment method, customer count by tenure group, monthly charges by churn, customer distribution by contract and gender, tenure versus monthly charges, and customer details.

Interactive slicers are used for Gender, Contract, Internet Service, Payment Method, Senior Citizen and Churn so the dashboard can be filtered without creating multiple report pages.

Core DAX Measures

Total Customers =
DISTINCTCOUNT(Customer_Churn_Cleaned[Customer_ID])

Churned Customers =
CALCULATE(
DISTINCTCOUNT(Customer_Churn_Cleaned[Customer_ID]),
Customer_Churn_Cleaned[Churn] = "Yes"
)

Retained Customers =
CALCULATE(
DISTINCTCOUNT(Customer_Churn_Cleaned[Customer_ID]),
Customer_Churn_Cleaned[Churn] = "No"
)

Churn Rate % =
DIVIDE([Churned Customers], [Total Customers], 0)

Average Monthly Charges =
AVERAGE(Customer_Churn_Cleaned[Monthly_Charges])

Average Tenure =
AVERAGE(Customer_Churn_Cleaned[Tenure_Months])

5. Business Insights

1. The overall churn rate is **41.75%**: 501 of 1200 customers churned, while 699 were retained.
2. **Month-to-month contracts** have the highest churn rate at **59.51%**, compared with 25.42% for one-year and 15.89% for two-year contracts.
3. **Electronic check** has the highest payment-method churn rate at **51.24%**. The other payment methods are substantially lower, around 35-37%.
4. **Fiber optic** customers have the highest internet-service churn rate at **53.58%**, versus 36.90% for DSL and 24.44% for customers with no internet service.
5. Customers in the **0-12 month tenure group** show the highest churn rate at **60.95%**, indicating that the earliest stage of the customer relationship requires attention.
6. Churned customers have higher average monthly charges (**84.88**) than retained customers (**74.57**). This supports an association between higher monthly charges and churn in this dataset.
7. The correlation between `Monthly_Charges` and `Churn_Flag` is **0.22**, a positive but modest relationship. `Tenure_Months` has a correlation of **-0.14**, indicating a weak negative linear relationship with churn.
8. Gender differences are relatively small: female churn is **43.05%** and male churn is **40.52%**. Contract, payment method and internet service show much larger differences.
9. The combination of **short tenure, month-to-month contracts, electronic-check payment and fiber-optic service** should be treated as a high-attention profile because each of these categories is individually associated with elevated churn in the analysis.
10. A practical retention focus would be early-tenure onboarding, encouraging suitable longer-term contracts, reviewing the electronic-check customer experience, and investigating the service/value experience of fiber-optic customers. These are analytical priorities suggested by association, not proof of causation.

6. Answers to Required Business Questions

1. **Overall churn rate:** 41.75%.
2. **Highest-churn contract:** Month-to-month (59.51%).
3. **Highest-churn payment method:** Electronic check (51.24%).
4. **Highest-churn internet service:** Fiber optic (53.58%).
5. **Is tenure related to churn?** Yes, descriptively. The 0-12 month group has the highest churn rate, although the linear correlation is weak (-0.14).
6. **Are higher monthly charges associated with churn?** Yes, descriptively. Churned customers average 84.88 versus 74.57 for retained customers; the correlation with `Churn_Flag` is 0.22.
7. **Highest-churn customer segment:** The strongest high-attention profile combines categories that individually show elevated churn: early tenure, month-to-month contract, electronic check and fiber optic service.
8. **Major associated factors:** Contract type, payment method, internet service, tenure group and monthly charges.
9. **Segment requiring attention:** Early-tenure month-to-month customers, particularly those using electronic check and fiber optic service.

7. Conclusion

The project demonstrates an end-to-end analytics workflow: data loading, exploration, cleaning, feature creation, descriptive analysis, visualization and interactive dashboarding. The strongest churn patterns appear around flexible month-to-month contracts, electronic-check payments, fiber-optic service, early tenure and higher monthly charges. These findings can guide further investigation and targeted retention analysis.

8. Deliverables

Submission package: cleaned CSV dataset, Python analysis script, Power BI dashboard file, PDF project report, README file and a folder containing the generated Python visualizations.